8. Three fair coins are tossed together. Find the probability of getting

(i) all heads (ii) atleast one tail

(iii) atmost one head (iv) atmost two tails

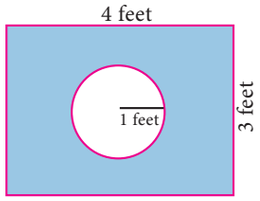
9. A bag contains 5 red balls, 6 white balls, 7 green balls, 8 black balls. One ball is drawn at random from the bag. Find the probability that the ball drawn is

(i) white (ii) black or red

(iii) not white (iv) neither white nor black

10. In a box there are 20 non-defective and some defective bulbs. If the probability that a bulb selected at random from the box found to be defective is $\frac{3}{8}$ then, find the number of defective bulbs.

11. Some boys are playing a game, in which the stone thrown by them landing in a circular region (given in the figure) is considered as win and landing other than the circular region is considered as loss. What is the probability to win the game? ($\pi = 3.14$)



12. Two customers Priya and Amuthan are visiting a particular shop in the same week (Monday to Saturday). Each is equally likely to visit the shop on any one day as on another day. What is the probability that both will visit the shop on

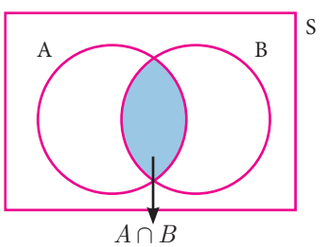
(i) the same day (ii) different days (iii) consecutive days?

13. In a game, the entry fee is ₹150. The game consists of tossing a coin 3 times. Dhana bought a ticket for entry . If one or two heads show, she gets her entry fee back. If she throws 3 heads, she receives double the entry fees. Otherwise she will lose. Find the probability that she (i) gets double entry fee (ii) just gets her entry fee (iii) loses the entry fee.

8.5 Algebra of Events

In a random experiment, let S be the sample space. Let $A \subseteq S$ and $B \subseteq S$ be the events in S. We say that

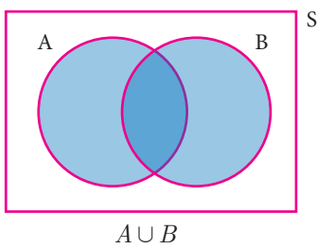
(i) $(A \cap B)$ is an event that occurs only when both A and B occurs.



$A \cap B$

Fig. 8.6(a)

(ii) $(A \cup B)$ is an event that occurs when either one of A or B occurs.



$A \cup B$

Fig. 8.6(b)

(iii) $\overline{A}$ is an event that occurs only when A doesn't occur.

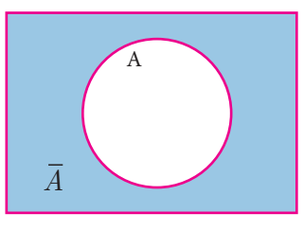


Fig. 8.6(c)

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